

Novel acetylcysteine regimens for treatment of paracetamol overdose.

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Paracetamol (acetaminophen) overdose remains the leading cause of death or transplantation due to acute liver failure in many parts of the world. Acetylcysteine has long been recognized as an effective antidote, via oral or intravenous administration, minimizing the risk and severity of acute liver injury if administered sufficiently early after a paracetamol overdose. Despite this, its mechanisms of action remain obscure, and there is uncertainty regarding the optimal dose and duration of treatment. The intravenous infusion protocol was originally developed as a three-step loading regimen; it causes very high early peak plasma concentrations of acetylcysteine whereas the later maintenance infusion is associated with much lower concentrations. This pharmacokinetic profile is associated with two particular concerns: a high rate of occurrence of adverse effects that occur after the initial loading infusion, and the possibility that the maintenance phase of treatment might deliver too low a dose of acetylcysteine for optimum protection against liver injury. Recently described novel administration regimens offer different rates of intravenous acetylcysteine administration in both the loading and maintenance phases. These alternative regimens appear to be well tolerated in small patient groups, but too few clinical data are available to evaluate their comparative efficacy in preventing paracetamol-induced liver injury.

N-acetylcysteine amide, a promising antidote for acetaminophen toxicity.

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Acetaminophen (N-acetyl-p-aminophenol, APAP) is one of the most widely used over the counter antipyretic and analgesic medications. It is safe at therapeutic doses, but its overdose can result in severe hepatotoxicity, a leading cause of drug-induced acute liver failure in the USA. Depletion of glutathione (GSH) is one of the initiating steps in APAP-induced hepatotoxicity; therefore, one strategy for restricting organ damage is to restore GSH levels by using GSH prodrugs. N-acetylcysteine (NAC), a GSH precursor, is the only currently approved antidote for an acetaminophen overdose. Unfortunately, fairly high doses and longer treatment times are required due to its poor bioavailability. In addition, oral and I.V. administration of NAC in a hospital setting are laborious and costly. Therefore, we studied the protective effects of N-acetylcysteine amide (NACA), a novel antioxidant with higher bioavailability, and compared it with NAC in APAP-induced hepatotoxicity in C57BL/6 mice. Our results showed that NACA is better than NAC at a low dose (106mg/kg) in preventing oxidative stress and protecting against APAP-induced damage. NACA significantly increased GSH levels and the GSH/GSSG ratio in the liver to 66.5% and 60.5% of the control, respectively; and it reduced the level of ALT by 30%. However, at the dose used, NAC was not effective in combating the oxidative stress induced by APAP. Thus, NACA appears to be better than NAC in reducing the oxidative stress induced by APAP. It would be of great value in the health care field to develop drugs like NACA as more effective and safer options for the prevention and therapeutic intervention in APAP-induced toxicity.

Cost minimization analysis comparing enteral N-acetylcysteine to intravenous acetylcysteine in the management of acute acetaminophen toxicity.

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Acetaminophen poisoning is one of the most common exposures and causes of poisoning-related fatalities as reported to U.S. poison information centers. Acetylcysteine is indicated for the antidotal treatment of acetaminophen poisoning to prevent or minimize acetaminophen-related hepatotoxicity. Available as either an enteral or intravenous (IV) formulation, both forms of acetylcysteine have been proven to be efficacious. Because of the differences in the acquisition costs and the length of treatment, it is unclear which treatment route is the most cost-effective. The purpose of this study was to compare the total hospitalization charges associated with patients who received either enteral or IV acetylcysteine therapy. A retrospective, IRB-approved cohort study of patients treated with either enteral or IV acetylcysteine at a university-related hospital for the treatment of acute acetaminophen overdose was conducted. Patients included were over 18 years of age, admitted during the 5-year periods of 1996-2000 (enteral) and 2004-2008 (IV), had an ICD-9 discharge diagnosis for acetaminophen overdose, had no transplant history, and were admitted within 24 h of the overdose. The primary endpoint was the total cost associated with the hospital stay. The Consumer Price Index (CPI) inflation calculator from the U.S. Bureau of Labor Statistics was used to adjust all monetary values to 2008 dollars. Of a total of 1,647 patients, 261 met the inclusion criteria with 70 patients being treated with enteral acetylcysteine and 191 patients treated with IV acetylcysteine. The associated cost was greater in the enteral group than in the IV group (\$18,287.63 vs. \$7,607.82; $p < 0.001$). The average length of stay was longer in the enteral group compared to the IV group (7 days vs. 4 days; $p < 0.001$). Patients who were treated with IV acetylcysteine had a decreased length of stay and cost of hospitalization compared with those patients who were treated with enteral acetylcysteine.

A dosing regimen for immediate N-acetylcysteine treatment for acute paracetamol overdose.

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Current treatment of paracetamol (acetaminophen) poisoning involves initiating a 3-phase N-acetylcysteine (NAC) infusion after comparing a plasma concentration, taken ≥ 4 h post-overdose, to a nomogram. This may result in dosing errors, a delay in treatment, or possibly more adverse effects - due to the use of a high dose rate for the first infusion when treatment is initiated. Our aim was to investigate a novel dosing regimen for the immediate administration of NAC on admission at a lower infusion rate. We used a published population pharmacokinetic model of NAC to simulate a scenario where a patient presents to the hospital 2 h post-overdose. The conventional regimen is commenced 6 h post-overdose when the 4-h plasma paracetamol concentration is available. We investigated an NAC infusion using a lower dosing rate initiated immediately on presentation. We determined a dosing rate that gave an area under the curve (AUC) of the concentration-time curve that was the same or greater than that from the conventional regimen on 90% of occasions. Lower dosing rates of NAC initiated immediately resulted in a similar exposure to NAC. An infusion of 110 mg/kg over the first 5 h (22 mg/kg/h) followed by the last two phases of the conventional regimen, or 200 mg/kg over 9 h (22.6 mg/kg/h) followed by the last phase of the conventional regimen could be used. The novel dosing regimen allowed immediate treatment of a patient using a lower dosing rate. This greatly simplifies the current dosing regimen and may reduce NAC adverse effects while ensuring the same amount of NAC is delivered.

An analysis of N-acetylcysteine treatment for acetaminophen overdose using a systems

model of drug-induced liver injury.

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N-acetylcysteine (NAC) is the treatment of choice for acetaminophen poisoning; standard 72-h oral or 21-h intravenous protocols are most frequently used. There is controversy regarding which protocol is optimal and whether the full treatment course is always necessary. It would be challenging to address these questions in a clinical trial. We used DILIsym, a mechanistic simulation of drug-induced liver injury, to investigate optimal NAC treatment after a single acetaminophen overdose for an average patient and a sample population ($n = 957$). For patients presenting within 24 h of ingestion, we found that the oral NAC protocol preserves more hepatocytes than the 21-h intravenous protocol. In various modeled scenarios, we found that the 21-h NAC infusion is often too short, whereas the full 72-h oral course is often unnecessary. We found that there is generally a good correlation between the time taken to reach peak serum alanine aminotransferase (ALT) and the time taken to clear N-acetyl-p-benzoquinone imine (NAPQI) from the liver. We also found that the most frequently used treatment nomograms underestimate the risk for patients presenting within 8 h of overdose ingestion. $V(\max)$ for acetaminophen bioactivation to NAPQI was the most important variable in the model in determining interpatient differences in susceptibility. In conclusion, DILIsym predicts that the oral NAC treatment protocol, or an intravenous protocol with identical dosing, is superior to the 21-h intravenous protocol and ALT is the optimal available biomarker for discontinuation of the therapy. The modeling also suggests that modification of the current treatment nomograms should be considered.

Assessment of N-acetylcysteine use for acetaminophen overdose in the emergency department of a community teaching hospital: A pilot study.

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N-acetylcysteine (NAC) is the first-line treatment for acetaminophen (APAP) overdose. However, using NAC inappropriately is associated with an increased risk of adverse effects as well as a substantial increase in hospitalization and healthcare costs. This study aims to assess NAC utilization for acute APAP overdose in the emergency department of a community teaching hospital in Saudi Arabia. A retrospective chart review in which the patients initiated on an NAC secondary to acute APAP overdose at KSUMC during the period of June 2015 till November 2018 were included and assessed based on developed validated evident-based protocol for administering NAC for acute APAP ingestion. A total of 29 patients received NAC treatment for acute APAP overdose; 15 of which were adults, and 14 were pediatrics. Appropriate prescribing of NAC was observed in 14 (48.28%) patients, whereas NAC was inappropriately indicated for 15 (51.72%) patients; 9 of them were adults and 6 patients were pediatric. APAP-ingestion <150 mg/kg (<200 mg/kg in children) was the most common reason for inappropriate use ($n = 7$, 46.67%) followed by administering NAC <4 hours post-APAP ingestion ($n = 4$, 26.67%). Improper NAC administration appears to be a significant issue among patients with APAP overdose. The utilization of a protocol for the management of APAP overdose will reduce the unnecessary usage of NAC.

N-acetylcysteine overdose after acetaminophen poisoning.

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N-acetylcysteine (NAC) is used widely and effectively in oral and intravenous forms as a specific antidote for acetaminophen poisoning. Here we report a rare case of iatrogenic NAC overdose following an error in preparation of the solution, and describe its clinical symptoms. Laboratory results and are presented and examined. A 23-year-old alert female patient weighing 65 kg presented to the emergency ward with weakness, lethargy, extreme fatigue, nausea, and dizziness. She had normal arterial blood gas and vital signs. An excessive dosage of NAC over a short period of time can lead to hemolysis, thrombocytopenia, and acute renal failure in patients with normal glucose-6-phosphate dehydrogenase, and finally to death. Considering the similarity between some of the clinical symptoms of acetaminophen overdose and NAC overdose, it is vitally important for the administration phases and checking of the patient's symptoms to be carried out attentively and cautiously.

Late extensive intravenous administration of N-acetylcysteine can reverse hepatic failure in acetaminophen overdose.

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Acetaminophen is a commonly used analgesic and has been shown to be a main cause of drug-induced liver failure. N-acetylcysteine (NAC) should be employed as the antidote in case of acetaminophen poisoning within the first 8-10 hours. Oral administration of NAC is universally recommended and due to the adverse effects, the intravenous administration of the agent is reserved for patients with oral intolerance and severe complications. We here report an 18-year-old man with severe liver failure due to a huge ingestion of acetaminophen, who was taken into the Loghman Hakim Hospital Poison Center 72 hours after attempted suicide. Regarding the poor prognostic clues as his level of consciousness and impaired liver functions, an extensive intravenous regimen of NAC was started. The patient survived the condition with an additional intravenous administration of NAC past the first 72 hours of treatment. We discuss that even in late phases of intoxication; high-dose intravenous NAC can serve a substantial improvement.

Incidence of adverse drug reactions induced by N-acetylcysteine in patients with acetaminophen overdose.

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Intravenous N-acetylcysteine (IV-NAC) is widely recognized as the antidote of choice for acetaminophen overdose. However, its use is not without adverse drug reactions (ADR) that might affect therapeutic outcome or lead to treatment delay. The aim of this study was to investigate the type and incidence of ADR induced by IV-NAC in patients treated for acetaminophen overdose. This is a retrospective study of patients admitted to the hospital for acute acetaminophen overdose over a period of 4 years (1 January 2005 to 31 December 2008). The primary outcome of interest in this study was the occurrence of ADR during NAC administration. Pearson chi-square test or Fisher's exact test, student's t test, and Mann-Whitney U test were used in univariate analysis. SPSS 15 was used for data analysis. Two hundred and fifty five patients were studied. Different types of ADR were observed in 119 (46.7%) cases. Of those patients, 83 (69.7%) had been treated with IV-NAC versus 36 (30.3%) who had not ($p < .001$). The following ADR were significantly associated with IV-NAC administration: vomiting ($p = .001$), flushing ($p < .001$), rash ($p < .001$), pruritus ($p < .001$), chest pain ($p = .001$), bronchospasm ($p = .03$), coughing ($p =$

.01), headache ($p = .001$), dizziness ($p < .001$), convulsion ($p = .03$), and hypotension ($p = .001$). ADR were mild in 54 (43.2%), moderate in 17 (13.6%), and severe in 12 (9.6%) patients. There were no ADR in 42 (33.6%) patients. Comparative results of the characteristics of patients who reacted to IV-NAC and nonreactors showed that patients with ADR had no significant difference in age, gender, ethnicity, amount ingested, latency time, and acetaminophen level than nonreactors. ADR to IV-NAC were common among patients with acetaminophen overdose, but mostly minor and all reported adverse reactions were easily managed.

Delayed administration of N-acetylcysteine blunts recovery after an acetaminophen overdose unlike 4-methylpyrazole.

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N-acetylcysteine (NAC) is the only clinically approved antidote against acetaminophen (APAP) hepatotoxicity. Despite its efficacy in patients treated early after APAP overdose, NAC has been implicated in impairing liver recovery in mice. More recently, 4-methylpyrazole (4MP, Fomepizole) emerged as a potential antidote in the mouse APAP hepatotoxicity model. The objective of this manuscript was to verify the detrimental effect of NAC and its potential mechanism and assess whether 4MP has the same liability. C57BL/6J mice were treated with 300 mg/kg APAP; 9 h after APAP and every 12 h after that, the animals received either 100 mg/kg NAC or 184.5 mg/kg 4MP. At 24 or 48 h after APAP, parameters of liver injury, mitochondrial biogenesis and cell proliferation were evaluated. Delayed NAC treatment had no effect on APAP-induced liver injury at 24 h but reduced the decline of plasma ALT activities and prevented the shrinkage of the areas of necrosis at 48 h. This effect correlated with down-regulation of key activators of mitochondrial biogenesis (AMPK, PGC-1 α , Nrf1/2, TFAM) and reduced expression of Tom 20 (mitochondrial mass) and PCNA (cell proliferation). In contrast, 4MP attenuated liver injury at 24 h and promoted recovery at 48 h, which correlated with enhanced mitochondrial biogenesis and hepatocyte proliferation. In human hepatocytes, 4MP demonstrated higher efficacy in preventing cell death compared to NAC when treated at 18 h after APAP. Thus, due to the wider treatment window and lack of detrimental effects on recovery, it appears that at least in preclinical models, 4MP is superior to NAC as an antidote against APAP overdose.